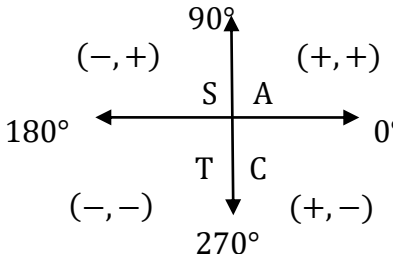


CALCULUS AND DIFFERENTIAL EQUATIONS (22MATS11)

Module 1 - Calculus

Prerequisites:

Trigonometry

Pythagorean identities						Reciprocal ratios					
$\sin^2 \theta + \cos^2 \theta = 1$						$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$					
$\sec^2 \theta - \tan^2 \theta = 1$						$\sec \theta = \frac{1}{\cos \theta}$					
$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$						$\cot \theta = \frac{1}{\tan \theta}$					
Sum formulas						Difference formulas					
$\sin(x + y) = \sin x \cos y + \cos x \sin y$						$\sin(x - y) = \sin x \cos y - \cos x \sin y$					
$\cos(x + y) = \cos x \cos y - \sin x \sin y$						$\cos(x - y) = \cos x \cos y + \sin x \sin y$					
$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$						$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$					
Double angle formulas						Triple angle formulas					
$\sin 2x = 2 \sin x \cos x$						$\sin 3x = 3 \sin x - 4 \sin^3 x$					
$\cos 2x = \cos^2 x - \sin^2 x$						$\cos 3x = 4 \cos^3 x - 3 \cos x$					
$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$						$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$					
Half angle formulas						Tangent formulas					
$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$						$\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$					
$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$						$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$					
$\tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$						$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$					
Standard angle formulas						ASTC Rule					
θ	0°	30°	45°	60°	90°						
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1						
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0						
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞						

Note:

$2 \sin^2 \frac{x}{2} = 1 - \cos x$	$\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)^2 = 1 + \sin x$	$\tan \left(\frac{\pi}{4} + x \right) = \frac{1 + \tan x}{1 - \tan x}$
$2 \cos^2 \frac{x}{2} = 1 + \cos x$	$\left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)^2 = 1 - \sin x$	$\tan \left(\frac{\pi}{4} - x \right) = \frac{1 - \tan x}{1 + \tan x}$

Same ratio formulas:

$\sin(-\theta) = -\sin \theta$	$\sin(2\pi - \theta) = -\sin \theta$	$\sin(\pi - \theta) = \sin \theta$	$\sin(\pi + \theta) = -\sin \theta$
$\cos(-\theta) = \cos \theta$	$\cos(2\pi - \theta) = \cos \theta$	$\cos(\pi - \theta) = -\cos \theta$	$\cos(\pi + \theta) = -\cos \theta$
$\tan(-\theta) = -\tan \theta$	$\tan(2\pi - \theta) = -\tan \theta$	$\tan(\pi - \theta) = -\tan \theta$	$\tan(\pi + \theta) = \tan \theta$
$\cot(-\theta) = -\cot \theta$	$\cot(2\pi - \theta) = -\cot \theta$	$\cot(\pi - \theta) = -\cot \theta$	$\cot(\pi + \theta) = \cot \theta$
$\sec(-\theta) = \sec \theta$	$\sec(2\pi - \theta) = \sec \theta$	$\sec(\pi - \theta) = -\sec \theta$	$\sec(\pi + \theta) = -\sec \theta$
$\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$	$\operatorname{cosec}(2\pi - \theta) = -\operatorname{cosec} \theta$	$\operatorname{cosec}(\pi - \theta) = \operatorname{cosec} \theta$	$\operatorname{cosec}(\pi + \theta) = -\operatorname{cosec} \theta$
(IV quadrant) Cos +ve	(IV quadrant) Cos +ve	(II quadrant) Sin +ve	(III quadrant) Tan +ve

Co ratio formulas:

$\sin \left(\frac{\pi}{2} - \theta \right) = \cos \theta$	$\sin \left(\frac{\pi}{2} + \theta \right) = \cos \theta$	$\sin \left(\frac{3\pi}{2} - \theta \right) = -\cos \theta$	$\sin \left(\frac{3\pi}{2} + \theta \right) = -\cos \theta$
$\cos \left(\frac{\pi}{2} - \theta \right) = \sin \theta$	$\cos \left(\frac{\pi}{2} + \theta \right) = -\sin \theta$	$\cos \left(\frac{3\pi}{2} - \theta \right) = -\sin \theta$	$\cos \left(\frac{3\pi}{2} + \theta \right) = \sin \theta$
$\tan \left(\frac{\pi}{2} - \theta \right) = \cot \theta$	$\tan \left(\frac{\pi}{2} + \theta \right) = -\cot \theta$	$\tan \left(\frac{3\pi}{2} - \theta \right) = \cot \theta$	$\tan \left(\frac{3\pi}{2} + \theta \right) = -\cot \theta$
$\cot \left(\frac{\pi}{2} - \theta \right) = \tan \theta$	$\cot \left(\frac{\pi}{2} + \theta \right) = -\tan \theta$	$\cot \left(\frac{3\pi}{2} - \theta \right) = \tan \theta$	$\cot \left(\frac{3\pi}{2} + \theta \right) = -\tan \theta$
$\sec \left(\frac{\pi}{2} - \theta \right) = \operatorname{cosec} \theta$	$\sec \left(\frac{\pi}{2} + \theta \right) = -\operatorname{cosec} \theta$	$\sec \left(\frac{3\pi}{2} - \theta \right) = -\operatorname{cosec} \theta$	$\sec \left(\frac{3\pi}{2} + \theta \right) = \operatorname{cosec} \theta$
$\operatorname{cosec} \left(\frac{\pi}{2} - \theta \right) = \sec \theta$	$\operatorname{cosec} \left(\frac{\pi}{2} + \theta \right) = \sec \theta$	$\operatorname{cosec} \left(\frac{3\pi}{2} - \theta \right) = -\sec \theta$	$\operatorname{cosec} \left(\frac{3\pi}{2} + \theta \right) = -\sec \theta$
(I quadrant) All +ve	(II quadrant) Sin +ve	(III quadrant) Tan +ve	(IV quadrant) Cos +ve

Differentiation of some standard functions

Non Trigonometric functions	Trigonometric functions	Hyperbolic functions	Inverse functions
$(k)' = 0$	$(\sin x)' = \cos x$	$(\sinh x)' = \cosh x$	$(\sin^{-1}x)' = \frac{1}{\sqrt{1-x^2}}$
$(x^n)' = n x^{n-1}$	$(\cos x)' = -\sin x$	$(\cosh x)' = \sinh x$	$(\cos^{-1}x)' = -\frac{1}{\sqrt{1-x^2}}$
$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$	$(\tan x)' = \sec^2 x$	$(\tanh x)' = \operatorname{sech}^2 x$	$(\tan^{-1}x)' = \frac{1}{1+x^2}$
$(\log x)' = \frac{1}{x}$	$(\cot x)' = -\operatorname{cosec}^2 x$	$(\coth x)' = -\operatorname{cosech}^2 x$	$(\cot^{-1}x)' = -\frac{1}{1+x^2}$
$(e^x)' = e^x$	$(\sec x)' = \sec x \cdot \tan x$	$(\operatorname{sech} x)' = -\operatorname{sech} x \cdot \tanh x$	$(\sec^{-1}x)' = \frac{1}{x\sqrt{x^2-1}}$
$(a^x)' = a^x \log a$	$(\operatorname{cosec} x)' = -\operatorname{cosec} x \cdot \cot x$	$(\operatorname{cosech} x)' = -\operatorname{cosech} x \cdot \coth x$	$(\operatorname{cosec}^{-1}x)' = -\frac{1}{x\sqrt{x^2-1}}$

Rules of differentiation

1. $(ku)' = ku'$	3. $(uv)' = uv' + vu'$
2. $(u \pm v)' = u' \pm v'$	4. $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$

1.1 Polar curves

Introduction:

- ❖ Polar coordinates are $(x, y) = (r \cos \theta, r \sin \theta)$ where r - radial distance, θ - polar angle.
- ❖ Polar form of the equation of the curve $r = f(\theta)$ is called polar curve.
- ❖ $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$, $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$, $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
- ❖ $\tan \left(\frac{\pi}{4} + \theta \right) = \frac{1 + \tan \theta}{1 - \tan \theta}$, $\tan \left(\frac{\pi}{4} - \theta \right) = \frac{1 - \tan \theta}{1 + \tan \theta}$
- ❖ Angle between radius vector and tangent is $\tan \phi = r \frac{d\theta}{dr}$

Problems:

1. Derive angle between radius vector and tangent. (May 22)

Let $P(r, \theta)$ be any point on the polar curve $r = f(\theta)$.

Let χ be the angle from the X axis to the tangent.

Let p be the perpendicular distance from the origin to the tangent.

By diagram, $\chi = \theta + \phi$

$$\tan \chi = \tan(\theta + \phi)$$

$$\tan \chi = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi} \text{ ----- (1)}$$

But

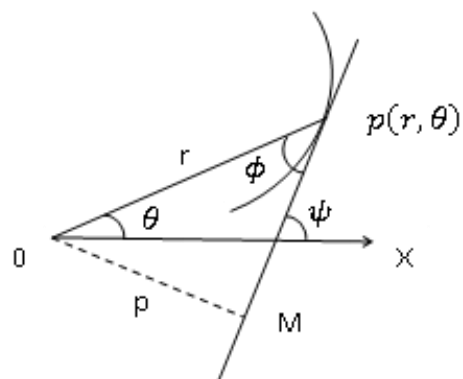
$$\tan \chi = \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta}(r \sin \theta)}{\frac{d}{d\theta}(r \cos \theta)} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

Divide by $\frac{dr}{d\theta} \cos \theta$ in numerator and denominator,

$$\tan \chi = \frac{\tan \theta + r \frac{d\theta}{dr}}{1 - r \frac{d\theta}{dr} \tan \theta} \text{ ----- (2)}$$

Equating components of (1) and (2),

$$\tan \phi = r \frac{d\theta}{dr}$$



2. Find the angle between radius vector and tangent to the following:

(i) $r^2 \cos 2\theta = a^2$ (ii) $r = a(1 + \cos \theta)$

(i) $r^2 \cos 2\theta = a^2$

Take log on both sides,

$$\log(r^2 \cos 2\theta) = \log a^2$$

$$\log r^2 + \log \cos 2\theta = 0$$

$$2 \log r + \log \cos 2\theta = 0$$

Differentiate with respect to θ ,

$$\frac{2}{r} \frac{dr}{d\theta} = \frac{2 \sin 2\theta}{\cos 2\theta}$$

$$\frac{1}{r} \frac{dr}{d\theta} = \tan 2\theta$$

$$\cot \phi = \cot \left(\frac{\pi}{2} - 2\theta \right)$$

$$\phi = \frac{\pi}{2} - 2\theta$$

(ii) $r = a(1 + \cos \theta)$

Take log on both sides,

$$\log r = \log a(1 + \cos \theta)$$

$$\log r = \log a + \log(1 + \cos \theta)$$

Differentiate with respect to θ ,

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{-\sin \theta}{(1 + \cos \theta)}$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{-2 \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\tan \frac{\theta}{2}$$

$$\cot \phi = \cot \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

At $\theta = \frac{\pi}{3}$,

$$\cot \phi = \cot \left(\frac{\pi}{2} + \frac{\pi}{6} \right)$$

Angle between the radius vector and

the tangent is $\phi = \frac{2\pi}{3}$

3. Find the angle between radius vector and tangent to the following:

(i) $r^n = a^n \sec(n\theta + \alpha)$ (ii) $r^m = a^m(\cos m\theta + \sin m\theta)$

(iii) $r^n = a^n \sec(n\theta + \alpha)$

Take log on both sides,

$$\log r^n = \log a^n + \log \sec(n\theta + \alpha)$$

$$n \log r = \log a^n + \log \sec(n\theta + \alpha)$$

Differentiate with respect to θ ,

$$\frac{n}{r} \frac{dr}{d\theta} = 0 + n \frac{\sec(n\theta + \alpha) \tan(n\theta + \alpha)}{\sec(n\theta + \alpha)}$$

$$\cot \phi = \tan(n\theta + \alpha)$$

$$\cot \phi = \cot\left(\frac{\pi}{2} - (n\theta + \alpha)\right)$$

Angle between the radius vector and the tangent is $\phi = \frac{\pi}{2} - n\theta - \alpha$

(iv) $r^m = a^m(\cos m\theta + \sin m\theta)$

Take log on both sides,

$$\log r^m = \log a^m(\cos m\theta + \sin m\theta)$$

$$\log r^m = \log a^m + \log(\cos m\theta + \sin m\theta)$$

Differentiate with respect to θ ,

$$\frac{m}{r} \frac{dr}{d\theta} = \frac{m(\cos m\theta - \sin m\theta)}{(\cos m\theta + \sin m\theta)}$$

$$\cot \phi = \tan\left(\frac{\pi}{4} - m\theta\right)$$

$$\cot \phi = \cot\left(\frac{\pi}{2} - \left(\frac{\pi}{4} - m\theta\right)\right)$$

Angle between the radius vector and the tangent is $\phi = \frac{\pi}{4} + m\theta$

4. Find the angle between radius vector and tangent to the following:

$$\frac{l}{r} = 1 + e \cos \theta.$$

$$\frac{l}{r} = 1 + e \cos \theta$$

Take log on both sides,

$$\log l - \log r = \log(1 + e \cos \theta)$$

Differentiate with respect to θ ,

$$-\frac{1}{r} \frac{dr}{d\theta} = \frac{-e \sin \theta}{1 + e \cos \theta}$$

$$\cot \phi = \frac{e \sin \theta}{1 + e \cos \theta}$$

$$\tan \phi = \frac{1 + e \cos \theta}{e \sin \theta}, \quad \phi = \tan^{-1}\left(\frac{1 + e \cos \theta}{e \sin \theta}\right)$$

5. Show that the following pair of curves intersect orthogonally:

(i) $r = a(1 + \cos \theta)$, $r = b(1 - \cos \theta)$

$r = a(1 + \cos \theta)$ Take log on both sides, $\log r = \log a + \log(1 + \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin \theta}{1 + \cos \theta}$ $\frac{1}{r} \frac{dr}{d\theta} = \frac{-2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}$ $\cot \phi_1 = -\tan \frac{\theta}{2}$	$r = b(1 - \cos \theta)$ Take log on both sides, $\log r = \log b + \log(1 - \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$ $\frac{1}{r} \frac{dr}{d\theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}}$ $\cot \phi_2 = \cot \frac{\theta}{2}$
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Since $\cot \phi_1 \cdot \cot \phi_2 = -1$, both intersect orthogonally.

(ii) $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$

$r^n = a^n \cos n\theta$ Take log on both sides, $n \log r = n \log a + \log \cos n\theta$ Differentiate w. r. to θ $\frac{n}{r} \frac{dr}{d\theta} = \frac{-n \sin n\theta}{\cos n\theta} = -n \tan n\theta$ $\frac{1}{r} \frac{dr}{d\theta} = -\tan n\theta$ $\cot \phi_1 = -\tan n\theta$	$r^n = b^n \sin n\theta$ Take log on both sides, $\log r = n \log b + \log \sin n\theta$ Differentiate w. r. to θ $\frac{n}{r} \frac{dr}{d\theta} = \frac{n \sin n\theta}{\cos n\theta} = n \cot n\theta$ $\frac{1}{r} \frac{dr}{d\theta} = \cot n\theta$ $\cot \phi_2 = \cot n\theta$
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Since $\cot \phi_1 \cdot \cot \phi_2 = -1$, both intersect orthogonally.

(iii) $r = \frac{a}{1+\cos \theta}$ and $r = \frac{b}{1-\cos \theta}$

$r = \frac{a}{1+\cos \theta}$ Take log on both sides, $\log r = \log a - \log(1 + \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1+\cos \theta} = \tan \frac{\theta}{2}$ $\cot \phi_1 = \tan \frac{\theta}{2}$	$r = \frac{b}{1-\cos \theta}$ Take log on both sides, $\log r = \log b - \log(1 - \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin \theta}{1-\cos \theta} = -\cot \frac{\theta}{2}$ $\cot \phi_2 = -\cot \frac{\theta}{2}$
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Since $\cot \phi_1 \cdot \cot \phi_2 = -1$, both intersect orthogonally

(iv) $r = a\theta$ and $r = \frac{a}{\theta}$

$r = a\theta$ Take log on both sides, $\log r = \log a\theta$ $\log r = \log a + \log \theta$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{1}{\theta}$ $\cot \phi_1 = \frac{1}{\theta}$	$r = \frac{a}{\theta}$ Take log on both sides, $\log r = \log \frac{a}{\theta}$ $\log r = \log a - \log \theta$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = -\frac{1}{\theta}$ $\cot \phi_2 = -\frac{1}{\theta}$
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But $a\theta = \frac{a}{\theta} \Rightarrow \theta^2 = 1$. Since $\cot \phi_1 \cdot \cot \phi_2 = -\frac{1}{\theta^2} = -1$, both intersect orthogonally.

(v) $r = ae^\theta$ and $re^\theta = b$

$r = ae^\theta$ Take log on both sides, $\log r = \log a + \theta$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = 1$ $\cot \phi_1 = 1$	$re^\theta = a$ Take log on both sides, $\log r + \theta = \log a$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = -1$ $\cot \phi_2 = -1$
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Since $\cot \phi_1 \cdot \cot \phi_2 = -1$, both intersect orthogonally.

(vi) Show that $r = 4 \sec^2 \frac{\theta}{2}$ and $r = 9 \operatorname{cosec}^2 \frac{\theta}{2}$ the pair of curves cut orthogonally.

(May 22)

$r = 4 \sec^2 \frac{\theta}{2}$ Take log on both sides, $\log r = \log 4 + 2 \log \sec \frac{\theta}{2}$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{2}{\sec^2 \frac{\theta}{2}} \sec \frac{\theta}{2} \tan \frac{\theta}{2}$ $\cot \phi_1 = 2 \tan \frac{\theta}{2}$	$r = 9 \operatorname{cosec}^2 \frac{\theta}{2}$ Take log on both sides, $\log r = \log 9 + 2 \log \operatorname{cosec} \frac{\theta}{2}$ Differentiate with respect to θ $\frac{1}{r} \frac{dr}{d\theta} = -\frac{2}{\operatorname{cosec}^2 \frac{\theta}{2}} \operatorname{cosec} \frac{\theta}{2} \cot \frac{\theta}{2}$ $\cot \phi_2 = -2 \cot \frac{\theta}{2}$
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Since $\cot \phi_1 \cdot \cot \phi_2 = -1$, both intersect orthogonally.

6. Find the angle of intersection of the following pair of curves:

(i) $r = \sin \theta + \cos \theta$ and $r = 2 \sin \theta$

$r = \sin \theta + \cos \theta$ Take log on both sides $\log r = \log (\sin \theta + \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta}$ $\cot \phi_1 = \tan \left(\frac{\pi}{4} - \theta \right) = \cot \left(\frac{\pi}{4} + \theta \right)$ $\phi_1 = \frac{\pi}{4} + \theta$	$r = 2 \sin \theta$ Take log on both sides $\log r = \log(2 \sin \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{2 \cos \theta}{2 \sin \theta}$ $\cot \phi_2 = \cot \theta$ $\phi_2 = \theta$
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Therefore, $|\phi_1 - \phi_2| = \frac{\pi}{4}$

(ii) $r = a(1 - \cos \theta)$ and $r = 2a \cos \theta$

$r = a(1 - \cos \theta)$ Take log on both sides $\log r = \log a + \log (1 - \cos \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$ $\cot \phi_1 = \cot \frac{\theta}{2}$ $\phi_1 = \frac{\theta}{2}$	$r = 2a \cos \theta$ Take log on both sides $\log r = \log 2a + \log \cos \theta$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin \theta}{\cos \theta} = -\tan \theta$ $\cot \phi_2 = \cot \left(\frac{\pi}{2} + \theta \right)$ $\phi_2 = \frac{\pi}{2} + \theta$
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But By data, $1 - \cos \theta = 2 \cos \theta \Rightarrow \theta = \cos^{-1} \frac{1}{3}$

Therefore, $|\phi_1 - \phi_2| = \frac{\pi}{2} + \frac{\theta}{2} = \frac{\pi}{2} + \frac{1}{2} \cos^{-1} \left(\frac{1}{3} \right)$

(iii) $r = a \log \theta$ and $r = \frac{a}{\log \theta}$ (May 22)

$r = a \log \theta$ Take log on both sides $\log r = \log a + \log (\log \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{1}{\theta \log \theta}$ $\cot \phi_1 = \frac{1}{\theta \log \theta}$ $\tan \phi_1 = \theta \log \theta$	$r = \frac{a}{\log \theta}$ Take log on both sides $\log r = \log a - \log (\log \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = -\frac{1}{\theta \log \theta}$ $\cot \phi_2 = -\frac{1}{\theta \log \theta}$ $\tan \phi_2 = -\theta \log \theta$
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But by data, $a \log \theta = \frac{a}{\log \theta} \Rightarrow \theta = e$.

Therefore, $\tan(\phi_1 - \phi_2) = \frac{\tan \phi_1 - \tan \phi_2}{1 + \tan \phi_1 \tan \phi_2} = \frac{e + e}{1 + e^2} = \frac{2e}{1 + e^2}$

Therefore, $|\phi_1 - \phi_2| = \tan^{-1} \frac{2e}{1 + e^2} = 2 \tan^{-1} e$

(iv) $r = a \sin 2\theta$ and $r = a \cos 2\theta$

$r = a \sin 2\theta$ Take log on both sides $\log r = \log a + \log (\sin 2\theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = 2 \cot 2\theta$ $\cot \phi_1 = 2 \cot 2\theta$ $\tan \phi_1 = \frac{1}{2} \tan 2\theta$	$r = a \cos 2\theta$ Take log on both sides $\log r = \log a + \log (\cos 2\theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = -2 \tan 2\theta$ $\cot \phi_2 = -2 \tan 2\theta$ $\tan \phi_2 = -\frac{1}{2} \cot 2\theta$
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But by data, $a \sin 2\theta = a \cos 2\theta \Rightarrow 2\theta = \pi/4$

Therefore, $\tan|\phi_1 - \phi_2| = \left| \frac{\tan\phi_1 - \tan\phi_2}{1 + \tan\phi_1 \cdot \tan\phi_2} \right| = \left| \frac{\frac{1}{2} + \frac{1}{2}}{1 - \frac{1}{2} \cdot \frac{1}{2}} \right| = \frac{4}{3}$

Therefore, $|\phi_1 - \phi_2| = \tan^{-1} \frac{4}{3}$

(v) $r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$

$r = \frac{a\theta}{1+\theta}$ Take log on both sides $\log r = \log a\theta - \log (1 + \theta)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = \frac{1}{a\theta} (a) - \frac{1}{1 + \theta}$ $\cot \phi_1 = \frac{1}{\theta} - \frac{1}{1+\theta} = \frac{1}{\theta(1+\theta)}$ $\tan \phi_1 = \theta(1 + \theta)$	$r = \frac{a}{1+\theta^2}$ Take log on both sides $\log r = \log a - \log (1 + \theta^2)$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = - \frac{2\theta}{1+\theta^2}$ $\cot \phi_2 = - \frac{2\theta}{1+\theta^2}$ $\tan \phi_2 = - \frac{1+\theta^2}{2\theta}$
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By data, $\frac{a\theta}{1+\theta} = \frac{a}{1+\theta^2}$, $\theta + \theta^3 = 1 + \theta$, Therefore, $\theta = 1$.

$\tan \phi_1 = 2, \tan \phi_2 = -1, \tan(\phi_1 - \phi_2) = \frac{2+1}{1+2} = 1$

$\phi_1 - \phi_2 = \frac{\pi}{4}$

(vi) $r^2 \sin 2\theta = 4$ and $r^2 = 16 \sin 2\theta$

$r^2 \sin 2\theta = 4$ Take log on both sides $\log r^2 + \log \sin 2\theta = \log 4$ Differentiate w. r. to θ $\frac{2}{r} \frac{dr}{d\theta} + \frac{2 \cos 2\theta}{\sin 2\theta} = 0$ $\cot \phi_1 = -\cot 2\theta$ $\cot \phi_1 = \cot(-2\theta)$ $\phi_1 = -2\theta$	$r^2 = 16 \sin 2\theta$ Take log on both sides $\log r^2 = \log 16 + \log \sin 2\theta$ Differentiate w. r. to θ $\frac{2}{r} \frac{dr}{d\theta} = 0 + \frac{2 \cos 2\theta}{\sin 2\theta}$ $\cot \phi_2 = \cot 2\theta$ $\phi_2 = 2\theta$
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By data, $16 \sin^2 2\theta = 4$, $\sin 2\theta = \frac{1}{2}$, $2\theta = \frac{\pi}{6}$, $\theta = \frac{\pi}{12}$

Therefore, $|\phi_1 - \phi_2| = 4\theta = 4\left(\frac{\pi}{12}\right) = \frac{\pi}{3}$

1.2 Pedal equations

Introduction:

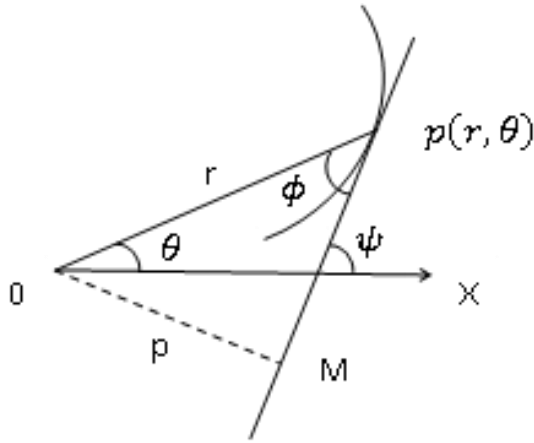
If p is the perpendicular distance from the pole to the tangent of the polar curve, then the equation of the curve in terms of p and r is called pedal equation or $p - r$ equation.

$p - r$ equation is $p = r \sin \phi$ or $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$.

Problems:

1. With usual notations, prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$ and hence deduce that

$$\frac{1}{p^2} = u^2 + u^4 \left(\frac{dr}{d\theta} \right)^2, \text{ where } u = \frac{1}{r}.$$



Let $P(r, \theta)$ – Any point on the polar curve $r = f(\theta)$.

Let r and p – Radius vector and perpendicular distance from the origin respectively.

By diagram, $\frac{p}{r} = \sin \phi \Rightarrow p = r \sin \phi$.

$$\frac{1}{p^2} = \frac{1}{r^2} \operatorname{cosec}^2 \phi = \frac{1}{r^2} (1 + \cot^2 \phi) = \frac{1}{r^2} \left(1 + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 \right) = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$$

$$\text{Therefore, } \frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$$

$$\text{If } u = \frac{1}{r}, \frac{du}{d\theta} = -\frac{1}{r^2} \left(\frac{dr}{d\theta} \right).$$

$$\frac{1}{p^2} = u^2 + \left(\frac{du}{d\theta} \right)^2$$

2. Find the pedal equation of the following curves:

(i) $r^2 = a^2 \sin^2 \theta$

To find: ϕ	To find: Pedal equation
$r^2 = a^2 \sin^2 \theta$	$p = r \sin \phi$
Take log on both sides,	$p = r \sin \theta$
$2 \log r = 2 \log a + 2 \log \sin \theta$	$p^2 = r^2 \sin^2 \theta$
$\log r = \log a + \log \sin \theta$	$p^2 = r^2 \left(\frac{r^2}{a^2} \right)$
$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta}{\sin \theta}$	$a^2 p^2 = r^4$
$\cot \phi = \cot \theta$	$ap = r^2$
$\phi = \theta$	

(ii) $r = 2(1 + \cos \theta)$

To find: ϕ	To find: Pedal equation
$r = 2(1 + \cos \theta)$	$p = r \sin \phi$
Take log on both sides,	$p = r \sin \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$
$\log r = \log 2 + \log(1 + \cos \theta)$	$p^2 = r^2 \cos^2 \frac{\theta}{2}$
$\log r = \log 2 + \log(1 + \cos \theta)$	$p^2 = r^2 \left(\frac{r}{4} \right)$
$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{-\sin \theta}{1 + \cos \theta} = \frac{-2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}$	$4p^2 = r^3$
$\cot \phi = -\tan \frac{\theta}{2} = \cot \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$	
$\phi = \frac{\pi}{2} + \frac{\theta}{2}$	

(iii) $r^n = a^n \cos n\theta$ (May 22)

To find: ϕ $r^n = a^n \cos n\theta$ Take log on both sides, $n \log r = n \log a + \log \cos n\theta$ Differentiate w. r. to θ $\frac{n}{r} \frac{dr}{d\theta} = 0 + \frac{-n \sin n\theta}{\cos n\theta}$ $\frac{1}{r} \frac{dr}{d\theta} = -\tan n\theta$ $\cot \phi = \cot \left(\frac{\pi}{2} + n\theta \right)$ $\phi = \frac{\pi}{2} + n\theta$	To find: Pedal equation $p = r \sin \phi$ $p = r \sin \left(\frac{\pi}{2} + n\theta \right)$ $p = r \cos n\theta$ $p = r \left(\frac{r^n}{a^n} \right)$ $a^n p = r^{n+1}$
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(iv) $r^m \cos m\theta = a^m$

To find: ϕ $r^m \cos m\theta = a^m$ Take log on both sides, $m \log r + \log \cos m\theta = m \log a$ Differentiate w. r. to θ $\frac{m}{r} \frac{dr}{d\theta} + \frac{-m \sin m\theta}{\cos m\theta} = 0$ $\frac{1}{r} \frac{dr}{d\theta} = \tan m\theta$ $\cot \phi = \cot \left(\frac{\pi}{2} - m\theta \right)$	To find: Pedal equation $p = r \sin \phi$ $p = r \sin \left(\frac{\pi}{2} - m\theta \right)$ $p = r \cos m\theta$ $p = r \left(\frac{a^m}{r^m} \right)$ $r^{m-1} p = a^m$
--	---

$\phi = \frac{\pi}{2} - m\theta$	
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(v) $r^m = a^m(\cos m\theta + \sin m\theta)$

To find: ϕ $r^m = a^m(\cos m\theta + \sin m\theta)$ Take log on both sides, $m \log r = \log a^m + \log(\cos m\theta + \sin m\theta)$ Differentiate w. r. to θ $\frac{m}{r} \frac{dr}{d\theta} = 0 + \frac{-m \sin m\theta + m \cos m\theta}{\cos m\theta + \sin m\theta}$ $\frac{1}{r} \frac{dr}{d\theta} = \tan\left(\frac{\pi}{4} - m\theta\right)$ $\cot \phi = \cot\left(\frac{\pi}{2} - \frac{\pi}{4} + m\theta\right)$ $\phi = \frac{\pi}{4} + m\theta$	To find: Pedal equation $p = r \sin \phi$ $p = r \sin\left(\frac{\pi}{4} + m\theta\right)$ $p = \frac{r}{\sqrt{2}}(\cos m\theta + \sin m\theta)$ $p = \frac{r}{\sqrt{2}}\left(\frac{r^m}{a^m}\right)$ $\sqrt{2}a^m p = r^{m+1}$
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3. Find the Pedal equation of the following curves:

(i) $r = ae^{m\theta}$

To find: ϕ $r = ae^{m\theta}$ Take log on both sides, $\log r = \log a + m\theta$ Differentiate w. r. to θ $\frac{1}{r} \frac{dr}{d\theta} = 0 + m$ $\cot \phi = m$	To find: Pedal equation $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta}\right)^2$ $\frac{1}{p^2} = \frac{1}{r^2} (1 + m^2)$ $r^2 = p^2 (1 + m^2)$
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(ii) $\frac{l}{r} = 1 + e \cos \theta$

To find: ϕ $\frac{l}{r} = 1 + e \cos \theta$ Take log on both sides, $\log l - \log r = \log(1 + e \cos \theta)$ Differentiate w. r. to θ $0 - \frac{1}{r} \frac{dr}{d\theta} = \frac{-e \sin \theta}{1 + e \cos \theta}$ $\cot \phi = \frac{e \sin \theta}{1 + e \cos \theta}$	To find: Pedal equation $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$ $\frac{1}{p^2} = \frac{1}{r^2} \left(1 + \frac{e^2 \sin^2 \theta}{(1 + e \cos \theta)^2} \right)$ $r^2(1 + e \cos \theta)^2 = p^2(1 + e^2 + 2e \cos \theta)$ $l^2 = p^2 \left(1 + e^2 + 2 \left(\frac{l}{r} - 1 \right) \right)$ $l^2 = p^2 \left(e^2 + \frac{2l}{r} - 1 \right)$
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(iii) $\frac{2a}{r} = 1 - \cos \theta$

To find: ϕ $\frac{2a}{r} = 1 - \cos \theta$ Take log on both sides, $\log 2a - \log r = \log(1 - \cos \theta)$ Differentiate w. r. to θ $0 - \frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}}$ $\cot \phi = -\cot \frac{\theta}{2} = \cot \left(\pi - \frac{\theta}{2} \right)$ $\phi = \pi - \frac{\theta}{2}$	To find: Pedal equation $p = r \sin \phi$ $p = r \sin \left(\pi - \frac{\theta}{2} \right) = r \sin \frac{\theta}{2}$ $\frac{p^2}{r^2} = \sin^2 \frac{\theta}{2} = \frac{a}{r}$ $p^2 = ar$
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1.3 Radius of curvature

Introduction:

The reciprocal of the curvature of a curve at any point p is called the radius of curvature at p.

It is defined by $\rho = \frac{ds}{d\psi}$.

Cartesian form	Polar form
$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$	$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}$
Parametric form	Pedal form
$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x' y'' - x'' y'}$	$\rho = r \frac{dr}{dp}$

1. Derive radius of curvature for the Cartesian curve $y = f(x)$. (May 22)

$$\tan \psi = \frac{dy}{dx}$$

$$\tan \psi = y_1$$

$$\psi = \tan^{-1}(y_1)$$

Differentiating w. r. to x ,

$$\frac{d\psi}{dx} = \frac{1}{1+y_1^2} \cdot y_2$$

Therefore, radius of curvature is given by

$$\rho = \frac{ds}{d\psi} = \frac{ds}{dx} \cdot \frac{dx}{d\psi} = \sqrt{1 + y_1^2} \cdot \frac{1+y_1^2}{y_2}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

2. Derive radius of curvature for the parametric curve $x = f(t), y = g(t)$.

$y_1 = \frac{dy}{dx}$ $= \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ $= \frac{y'}{x'}$	$y_2 = \frac{d}{dx} \left(\frac{y'}{x'} \right)$ $= \frac{d}{dt} \left(\frac{y'}{x'} \right) \left(\frac{dt}{dx} \right)$ $= \left(\frac{x' y'' - x'' y'}{x'^2} \right) \left(\frac{1}{x'} \right)$
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Therefore, radius of curvature is given by

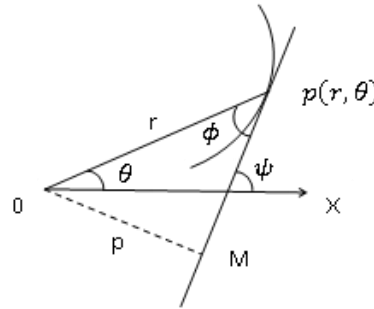
$$\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\left(1+\frac{y'^2}{x'^2}\right)^{\frac{3}{2}}}{\left(\frac{x' y'' - x'' y'}{x'^3}\right)}$$

$$\rho = \frac{(x'^2 + y'^2)^{\frac{3}{2}}}{x' y'' - x'' y'}$$

3. Derive radius of curvature for the polar curve $r = f(\theta)$.

By diagram, $\chi = \theta + \phi$

$$\begin{aligned} \frac{d\chi}{ds} &= \frac{d\theta}{ds} + \frac{d\phi}{ds} \\ &= \frac{d\theta}{ds} + \frac{d\phi}{d\theta} \cdot \frac{d\theta}{ds} \\ &= \frac{d\theta}{ds} \left(1 + \frac{d\phi}{d\theta} \right) \\ &= \frac{1 + \frac{d\phi}{d\theta}}{\frac{ds}{d\theta}} \text{----- (1)} \end{aligned}$$



But $\tan \phi = r \frac{d\theta}{dr}$

$$\phi = \tan^{-1} \left(\frac{r}{r_1} \right)$$

$$\frac{d\phi}{d\theta} = \frac{1}{1 + \left(\frac{r}{r_1}\right)^2} \cdot \frac{r_1 \cdot r_1 - r r_2}{r_1^2} = \frac{r_1^2 - r r_2}{r^2 + r_1^2}$$

$$1 + \frac{d\phi}{d\theta} = 1 + \frac{r_1^2 - r r_2}{r^2 + r_1^2} = \frac{r^2 + 2r_1^2 - r r_2}{r^2 + r_1^2}$$

$$\frac{ds}{d\theta} = \sqrt{r^2 + r_1^2}$$

Therefore, radius of curvature is given by

$$\begin{aligned} \frac{1}{\rho} &= \frac{d\chi}{ds} \\ &= \frac{1 + \frac{d\phi}{d\theta}}{\frac{ds}{d\theta}} \quad [\text{By (1)}] \\ &= \frac{1}{\sqrt{r^2 + r_1^2}} \cdot \frac{r^2 + 2r_1^2 - r r_2}{r^2 + r_1^2} \end{aligned}$$

$$\rho = \frac{(r^2 + r_1^2)^{\frac{3}{2}}}{r^2 + 2r_1^2 - r r_2}$$

Note:

$$\frac{ds}{dr} = \frac{ds}{d\theta} \cdot \frac{d\theta}{dr} = \left(\sqrt{r^2 + r_1^2} \right) \frac{1}{r_1} = \sqrt{r^2 \left(\frac{d\theta}{dr} \right)^2 + 1}$$

$$\frac{ds}{dr} = \sqrt{1 + \tan^2 \phi} = \sec \phi$$

$$\cos \phi = \frac{dr}{ds}$$

$$\sin \phi = \tan \phi \cdot \cos \phi = r \frac{d\theta}{dr} \cdot \frac{dr}{ds} = r \frac{d\theta}{ds}$$

4. Derive radius of curvature for the pedal curve $p = f(r)$.

By diagram, $\chi = \theta + \phi$

$$\frac{d\chi}{ds} = \frac{d\theta}{ds} + \frac{d\phi}{ds}$$

But $p = r \sin \phi$

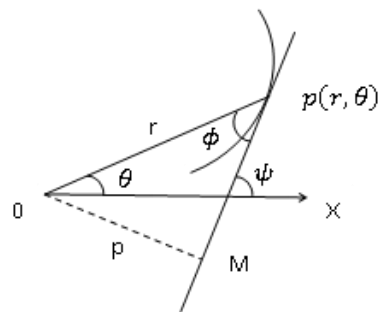
$$\frac{dp}{dr} = \sin \phi + r \cos \phi \frac{d\phi}{dr}$$

$$= r \frac{d\theta}{ds} + r \frac{dr}{ds} \cdot \frac{d\phi}{dr}$$

$$= r \left(\frac{d\theta}{ds} + \frac{d\phi}{ds} \right)$$

$$= r \left(\frac{d\chi}{ds} \right)$$

$$\rho = r \frac{dr}{dp}$$



5. Find the radius of curvature for $x^4 + y^4 = 2$ at $(1, 1)$ (July '16)

$$x^4 + y^4 = 2$$

Differentiate w. r. to x ,

$$4x^3 + 4y^3 y' = 0$$

$$x^3 + y^3 y' = 0 \text{ ----- (1)}$$

Differentiate again w. r. to x ,

$$3x^2 + 3y^2 (y')^2 + y^3 y'' = 0 \text{ ----- (2)}$$

At $(1, 1)$, $y' = -1$, $y'' = -6$.

Radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+1)^{3/2}}{-6} = \frac{2\sqrt{2}}{-6} = \frac{\sqrt{2}}{-3}$$

6. Find the radius of curvature of the Folium $x^3 + y^3 = 3axy$ at the point $(3a/2, 3a/2)$.

$$x^3 + y^3 = 3axy$$

Differentiate with respect to x

$$x^2 + y^2 y' = a(xy' + y) \text{ ----- (1)}$$

Differentiate again with respect to x

$$2x + 2y(y')^2 + y^2 y'' = a(xy'' + 2y') \text{ ----- (2)}$$

$$\text{At } \left(\frac{3a}{2}, \frac{3a}{2}\right),$$

$$(1) \Rightarrow \frac{9a^2}{4} + \frac{9a^2}{4} y' = a\left(\frac{3a}{2} y' + \frac{3a}{2}\right)$$

$$\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right) y' = \frac{3a^2}{2} - \frac{9a^2}{4}, \quad y' = -1$$

$$(2) \Rightarrow 2\left(\frac{3a}{2}\right) + 2\left(\frac{3a}{2}\right) + \left(\frac{3a}{2}\right)^2 y'' = a\left(\frac{3a}{2} y'' - 2\right)$$

$$3a + 3a + \frac{9a^2}{4} y'' - \frac{3a^2}{2} y'' = -2a$$

$$\frac{3a^2}{4} y'' = -8a, \quad y'' = \frac{-32}{3a}$$

Radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \left(\frac{2^2}{-32}\right) 3a = -\frac{3a\sqrt{2}}{16}$$

7. Find the radius of curvature of the catenary $y = c \cosh \frac{x}{c}$ at $(c, 0)$.

$$y = c \cosh \frac{x}{c}$$

Differentiate with respect to x ,

$$y_1 = \sinh \frac{x}{c} \text{ ----- (1)}$$

Differentiate again with respect to x ,

$$y_2 = \frac{1}{c} \cosh \frac{x}{c} \text{ ----- (2)}$$

$$\text{At } (c, 0), \quad y_1 = \sinh 1, \quad y_2 = \frac{1}{c} \cosh 1$$

Radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+\sinh^2 1)^{3/2}}{\frac{1}{c} \cosh 1} = c \cosh^2 1 = \frac{y^2}{c}$$

8. Find the radius of curvature of the parabola $y^2 = 4ax$ at $(at^2, 2at)$.

$$y^2 = 4ax$$

Differentiate w. r. to x,

$$yy_1 = 2a \text{ ----- (1)}$$

Differentiate again w. r. to x,

$$y_1^2 + yy_2 = 0 \text{ ----- (2)}$$

At $(at^2, 2at)$,

$$y_1 = \frac{1}{t}, \quad y_2 = -\frac{1}{2at^3}$$

Radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\left(1+\frac{1}{t^2}\right)^{3/2}}{\left(-\frac{1}{2at^3}\right)} = -2a(1+t^2)^{\frac{3}{2}}$$

9. Find the radius of curvature of the curve $y = x^3(x - a)$ at $(a, 0)$.

$$y = x^3(x - a)$$

Differentiate w. r. to x,

$$y_1 = x^3 - (x - a)3x^2 \text{ ----- (1)}$$

Differentiate again w. r. to x,

$$y_2 = 6x^2 + 6x(x - a) \text{ ----- (2)}$$

$$\text{at } (a, 0), \quad y_1 = a^3, \quad y_2 = 6a^2$$

Radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+a^6)^{\frac{3}{2}}}{6a^2}$$

10. For the cardioid $r = a(1 - \cos \theta)$, show that $\frac{\rho^2}{r}$ is a constant.

Step 1: Find ϕ $r = a(1 - \cos \theta)$ Take log on both sides, $\log r = \log a + \log(1 - \cos \theta)$ Differentiate w. r. to θ, $\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$ $\cot \phi = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}}$ $\cot \phi = \cot \frac{\theta}{2}$ $\phi = \frac{\theta}{2}$ Step 2: Find $p - r$ equation $p = r \sin \phi = r \sin \frac{\theta}{2}$ $p^2 = r^2 \sin^2 \frac{\theta}{2}$ $p^2 = \frac{r^3}{2a}$ $r^3 = 2ap^2$	Step 3: Find radius of curvature $r^3 = 2ap^2$ Differentiate w.r.to p $3r^2 \frac{dr}{dp} = 4ap$ $r \frac{dr}{dp} = \frac{4ap}{3r}$ $\rho = \frac{4ap}{3r}$ $\rho^2 = \frac{16a^2 p^2}{9r^2} = \frac{8ar}{9}$ Therefore, $\frac{\rho^2}{r} \text{ is a constant.}$
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11. Find the radius of curvature for $y^2 = \frac{a^2(a-x)}{x}$ where the curve meets the x-axis.

$$y^2 = \frac{a^2(a-x)}{x}$$

$$xy^2 = a^3 - a^2x \text{ ---- (1)}$$

Differentiate w.r.to x ,

$$2xyy' + y^2 = -a^2$$

$$2xyy' = -\frac{a^3}{x}$$

$$\frac{dy}{dx} = -\frac{a^3}{2yx^2} \text{ does not exist at } y = 0.$$

$$\therefore x_1 = \frac{dx}{dy} = -\frac{2x^2y}{a^3}$$

$$x_2 = \frac{d^2x}{dy^2} = -\frac{2}{a^3} \left(x^2 + 2xy \frac{dx}{dy} \right).$$

At $y = 0$,

$$x_1 = 0, x_2 = -\frac{2x^2}{a^3}$$

Therefore, radius of curvature is given by

$$\rho = \frac{(1+x_1^2)^{3/2}}{x_2} = \frac{-a^3}{2x^2}.$$

12. If ρ_1, ρ_2 be the radii of curvature at the extremities of any chord of the cardioid

$$r = a(1 + \cos\theta) \text{ which passes through the pole, show that } \rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$$

(May 22)

Step 1: Find ϕ

$$r = a(1 + \cos\theta)$$

$$\log r = \log a + \log(1 + \cos\theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin\theta}{1+\cos\theta}$$

$$\cot\phi = -\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}$$

$$= -\tan\frac{\theta}{2}$$

$$= \cot\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$$

$$\phi = \frac{\pi}{2} + \frac{\theta}{2}$$

Step 2: Find $p - r$ equation

$$p = r \sin\phi$$

$$= r \sin\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$$

$$p^2 = r^2 \cos^2\frac{\theta}{2}$$

$$p^2 = \frac{r^3}{2a} \left[\because \text{By data, } r = 2a \cos^2\frac{\theta}{2} \right]$$

$$r^3 = 2ap^2$$

Step 3: Find ρ

$$r^3 = 2ap^2$$

Differentiate w.r.to p

$$3r^2 \frac{dr}{dp} = 4ap$$

$$r \frac{dr}{dp} = \frac{4ap}{3r}$$

$$\rho = \frac{4ap}{3r}$$

$$\rho^2 = \frac{16a^2 p^2}{9r^2}$$

$$= \frac{8ar}{9}$$

Step 4: To prove $\rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$

$$\text{At } (r, \theta), \rho_1^2 = \frac{8ar}{9} = \frac{8a^2}{9} (1 + \cos\theta)$$

$$\text{At } (r, \pi + \theta), \rho_2^2 = \frac{8ar}{9} = \frac{8a^2}{9} (1 - \cos\theta).$$

$$\text{Adding both, } \rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$$

13. Show that the radius of curvature at any point of the cycloid

$$x = a(\theta + \sin \theta), \quad y = a(1 - \cos \theta) \text{ is } 4a \cos\left(\frac{\theta}{2}\right)$$

$$\frac{dx}{d\theta} = a(1 + \cos \theta), \quad \frac{dy}{d\theta} = a \sin \theta$$

$\begin{aligned} y_1 &= \frac{dy}{dx} \\ &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ &= \frac{a \sin \theta}{a(1 + \cos \theta)} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan \frac{\theta}{2} \end{aligned}$	$\begin{aligned} y_2 &= \frac{d^2y}{dx^2} \\ &= \frac{1}{2} \sec^2 \frac{\theta}{2} \times \frac{d\theta}{dx} \\ &= \frac{\frac{1}{2} \sec^2 \frac{\theta}{2}}{a(1 + \cos \theta)} \\ &= \frac{\frac{1}{2} \sec^2 \frac{\theta}{2}}{a(2 \cos^2 \frac{\theta}{2})} = \frac{1}{4a} \sec^4 \frac{\theta}{2} \end{aligned}$
---	---

Therefore, radius of curvature is given by

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\sec^3 \frac{\theta}{2}}{\frac{1}{4a} \sec^4 \frac{\theta}{2}} = 4a \cos \frac{\theta}{2}$$

14. Show that the radius of curvature at any point of the cycloid

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta) \text{ is } 4a \sin\left(\frac{\theta}{2}\right)$$

$$\frac{dx}{d\theta} = a(1 - \cos \theta), \quad \frac{dy}{d\theta} = a \sin \theta$$

$\begin{aligned} y_1 &= \frac{dy}{dx} \\ &= \frac{a \sin \theta}{a(1 - \cos \theta)} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} \\ &= \cot \frac{\theta}{2} \end{aligned}$	$\begin{aligned} y_2 &= \frac{d^2y}{dx^2} \\ &= -\frac{1}{2} \operatorname{cosec}^2\left(\frac{\theta}{2}\right) \times \frac{d\theta}{dx} \\ &= -\frac{1}{2} \operatorname{cosec}^2\left(\frac{\theta}{2}\right) \times \frac{1}{a(1 - \cos \theta)} \\ &= -\frac{1}{4a} \operatorname{cosec}^4\left(\frac{\theta}{2}\right) \end{aligned}$
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Therefore, radius of curvature is given by

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1 + \cot^2 \frac{\theta}{2})^{\frac{3}{2}}}{-\frac{1}{4a} \operatorname{cosec}^4(\frac{\theta}{2})} = -4a \sin \frac{\theta}{2}$$

Ignoring sign, $\rho = 4a \sin \frac{\theta}{2}$

15. Show that the radius of curvature at any point of the cycloid

$$x = a \cos^3 t, y = a \sin^3 t \text{ at } t = \frac{\pi}{4}.$$

$$\frac{dx}{d\theta} = -3a \cos^2 t \sin t, \quad \frac{dy}{d\theta} = 3a \sin^2 t \cos t$$

$y_1 = \frac{dy}{dx}$ $= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$ $= \frac{3a \sin^2 t \cos t}{-3a \cos^2 t \sin t}$ $= -\tan t$	$y_2 = \frac{d^2 y}{dx^2}$ $= -\sec^2 t \times \frac{dt}{dx}$ $= \frac{\sec^2 t}{3a \cos^2 t \sin t}$ $= \frac{1}{3a} \frac{\sec^4 t}{\sin t}$
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$$\text{At } t = \frac{\pi}{4}, y_1 = -1, y_2 = \frac{1}{3a} 4\sqrt{2}$$

Therefore, radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1+1)^{\frac{3}{2}}}{\frac{1}{3a} 4\sqrt{2}} = \frac{3a}{2}$$

16. Find the radius of curvature of the curve $x = a(\cos t + t \sin t), y = a(\sin t - t \cos t)$ at any point t .

$$\frac{dx}{dt} = a(-\sin t + \sin t + t \cos t) = at \cos t, \quad \frac{dy}{dt} = a(\cos t + t \sin t - \cos t) = at \sin t$$

$y_1 = \frac{dy}{dx}$ $= \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ $= \frac{at \sin t}{at \cos t}$ $= \tan t$	$y_2 = \frac{d^2 y}{dx^2}$ $= \sec^2 t \times \frac{dt}{dx}$ $= \frac{\sec^2 t}{at \cos t}$ $= \frac{1}{at} \sec^3 t$
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Therefore, radius of curvature is given by

$$\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1+\tan^2 t)^{\frac{3}{2}}}{\frac{1}{at} \sec^3 t} = \frac{\sec^3 t}{\frac{1}{at} \sec^3 t} = at.$$